Qualification Pack

Technician - Power System Transmission

QP Code: PSS/Q3401

Version: 1.0

NSQF Level: 4

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PSS/Q3401: Technician - Power System Transmission

Brief Job Description

The individual in the job role maintains sub-station, poles, towers and other related hardware in transmission system. He/she installs, maintains and repairs overhead and underground powerlines, cables and other associated equipment such as insulators, conductors, lightning arrestors, switches, metering systems, transformers and lighting systems.

Personal Attributes

The individual must be physically and mentally fit to be able to perform essential functions of the job safely. This will also include differently abled people who can perform the job with or without reasonable accommodations (modified practices.) The individual should be able to climb ladders, scaffolds, poles and towers of various heights. He/she should be able to work in confined spaces such as attics, manholes and crawlspaces. The individual should be able to read and understand written and verbal instructions respectively.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. PSS/N1331: Apply basic health and safety practices for power related work
2. PSS/N1336: Working effectively with others
3. PSS/N3401: Perform repair and maintenance of power transmission lines and components

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Transmission
Occupation	Operation and Maintenance - Transmission
Country	India
NSQF Level	4
Credits	NA
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7413.0100

Minimum Educational Qualification & Experience	10th Class with 2-3 Years of experience As a Technical helper/apprentice
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	Electrical - 6 months
Minimum Job Entry Age	18 Years
Last Reviewed On	11/03/2019
Next Review Date	11/03/2023
NSQC Approval Date	22/08/2019
Version	1.0
Reference code on NQR	2019/POW/PSSC/03474
NQR Version	1.0

PSS/N1331: Apply basic health and safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, assets and the environment

Scope

This unit/task covers the following:

- Follow workplace health and safety practices
- Follow fire safety practices
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Follow workplace health and safety practices

To be competent, the user/individual on the job must be able to:

PC1. • use protective clothing/equipment for specific tasks and work conditionsProtective Clothing • leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visorsequipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator

PC2. identify job-site hazards and possible causes of risks/accident at the workplaceHazards- electrical hazards; sharp edged and heavy tools; heated metals; oxyfuel and gas cylinders; welding radiation; hazardous surfaces; hazardous substances; physical hazards possible causes of risk and accident: physical actions; not following instructions; inattention; sickness and incapacity; health; not taking safety precautions

PC3. follow procedures for safe working with electrical equipmentSafe Working Procedures: tag out/lock out, ptw (permit to work)

PC4. follow warning signs while working with electrical systems Warning Signs: danger, high voltage area, out of service, etc

PC5. use standard safe working practices when working at heights, confined areas and trenches

PC6. test any electrical equipment and system using insulated testing devices before touching them

PC7. ensure positive isolation of electrical equipment & system as per the given standards

PC8. identify any abnormalities in electrical equipment or system installed alarm and/or noticing parameters from gauge/ indicator installedParameters: temperature, pressure, flow and current

PC9. follow safe working practices while working in hazardous conditions to ensure the safety of self and others Safe Working Practices: using ppe; installing and reading safety signs; handle tools in the correct manner and store and maintain them properly; keep work area clear of clutter; while working with electricity take all electrical precautions; safe lifting and carrying practices; use equipment that is working properly and is well maintained; take due measures for safety while working at heights, etc. take due measures for safety while working in confined spaces or trenches, etc.

PC10. use various methods of accident prevention in the work environmentMethods of Accident Prevention: training in health and safety procedures; using health and safety procedures; use of equipment and working practices; safety notices, advice; instruction from colleagues and supervisors

PC11. identify and retrieve general health and safety equipment in the workplaceGeneral Health and Safety Equipment: fire extinguishers; first aid equipment; safety instruments and clothing etc.

PC12. set up and use scaffolds, elevated platforms and ladders safelySet up: firm/level base, clip/lash down, leaning at the correct angle, appropriate load as per capacity, etc

PC13. inspect scaffolds, elevated platforms and ladders for faults and rectify themFaults: corrosion of metal components, deterioration, splits and cracks in timber components, imbalance, loose rungs, missing/ unfixed nuts or bolts, etc.

PC14. lift, carry and transport heavy objects & tools safely using correct procedures from storage to the workplace and vice versa

PC15. inspect power plant and its equipment routinely for any signs of oil, water and steam leakage

PC16. store flammable materials and machine lubricating oil safely and correctly

PC17. check that the emission and pollution control devices are working properly in line with environmental policy standards

PC18. use good housekeeping practices at all timesGood Housekee Practices: clean/tidy work area, removal/disposal of waste material, protect surfaces

PC19. identify common hazard signs displayed in various areasVarious Areas: chemical containers; equipment; packages; inside buildings; open areas and public spaces, etc.

PC20. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay

Follow fire safety practices

To be competent, the user/individual on the job must be able to:

PC21. use various appropriate fire extinguishers for different types of fires correctlyTypes of Fires: class a, b, c, d and e

PC22. use appropriate rescue techniques during fire hazard

PC23. maintain good housekeeping in order to prevent fire hazards

PC24. use a fire extinguisher correctly in case of fire

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC25. administer appropriate first aid to the victims for the relevant injuriesInjuries: bleeding, burns, choking, electric shock, poisoning etc.

PC26. act promptly and appropriately to an accident situation or medical emergency in real or simulated environments

PC27. perform and organise loss minimization or rescue activity during an accident in real or simulated environments

PC28. administer basic first aid including cpr to the victims of heart or cardiac arrest due to electric shock, before the arrival of emergency services

PC29. demonstrate basic emergency procedures during emergency mock drills for better understanding of procedures applied in emergenciesEmergency Procedures: raising alarm, safe/efficient evacuation, correct ways of escape, correct assembly point, roll call, correct return to work

PC30. free an affected person from electrocution safely as per the recommended procedure

PC31. move injured people and others to a safe place during emergencies

PC32. write an incident report or dictate a report to another person, and send report to the authorized personIncident Report Includes Details of: name, date/time of incident, date/time of report, location, environment conditions, persons involved, sequence of events, injuries sustained, damage sustained, actions taken, witnesses, supervisor/manager notified

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. names (and job titles if applicable), and where to find, all the people responsible for health and safety in a workplace
2. names and location of documents that refer to health and safety in the workplace
3. names and location of people responsible for health and safety in the workplace
4. documents that refer to health and safety in the workplacedocuments: fire notices, accident reports, safety instructions for equipment and procedures, company notices and documents, legal documents (e.g. government notices)
5. meaning of hazards and risks
6. health and safety hazards commonly present in the work environment and related precautions
7. possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible
8. possible causes of risk and accident possible causes of risk and accident: physical actions; not following instructions; inattention; sickness and incapacity (such as drunkenness); health hazards (such as untreated injuries and contagious illness); not taking safety precautions
9. methods of accident prevention
10. safe working practices when working with tools and machines
11. safe working practices while working at various hazardous sites
12. where to find all the general health and safety equipment in the workplace
13. various dangers associated with the use of electrical equipment
14. positive isolation of electrical equipment and system
15. safe handling and disposal of hazardous power plant wastes
16. use of emission and pollution control devices and measures taken to control pollution
17. various safety procedures and equipment used to work at heights, trenches and confined places
18. safe working practices specific to working with electrical equipment and system e.g. lock out/ tag out, ptw, etc.

19. preventative and remedial actions to be taken in the case of exposure to toxic materialstoxic materials: solvents, flux, lead exposure: ingested, contact with skin, inhaledpreventative action: ventilation, masks, protective clothing/ equipment); remedial action: immediate first aid, report to supervisor
20. importance of using protective clothing/equipment and other insulated work gear while handling electrical system and equipment
21. precautionary measures taken to prevent fire accident
22. various causes of firecauses of fires: heating of metal; spontaneous ignition; sparking; electrical heating; loose fires (smoking, welding, etc.); chemical fires; etc.
23. correct technique(s) of using a fire extinguisher
24. different methods of extinguishing fire
25. different materials used for extinguishing firematerials: sand, water, foam, co2, dry powder
26. emergency rescue techniques applied during a fire hazard
27. various types of safety signs and what they mean
28. appropriate basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries
29. rescue techniques applied during fire hazards
30. good housekeeping in order to prevent fire hazards
31. correct use of a fire extinguisher
32. how to free a person from electrocution
33. basic techniques of bandaging
34. perform artificial respiration and the cpr process

Generic Skills (GS)

User/individual on the job needs to know how to:

1. write an accident/ incident report in local language and/or english
2. read and comprehend basic content to read labels, charts, signages
3. read and comprehend basic english to read manuals of operation
4. read an accident/ incident report in local language or english
5. question coworkers appropriately in order to clarify instructions and other issues
6. give clear instructions to coworkers and subordinates
7. remain congenial while discussing and debating issues with co-workers
8. follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
9. ask for, provide and receive required assistance where possible to ensure achievement of work-related objectives
10. follow organizational rule-based decision-making process
11. take decisions with systematic course of actions and/or response
12. plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
13. build customer relationships and use customer centric approach

14. think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
15. seek appropriate assistance from other sources to resolve problems
16. report problems that cannot be resolved to the appropriate authority
17. identify cause and effect relations in own area of work
18. work systematically and logically to resolve the issues, identify the cause and anticipate unexpected results
19. critically evaluate the organizational processes to improve them

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Follow workplace health and safety practices	**19**	**43**	**-**	**-**
PC1. • use protective clothing/equipment for specific tasks and work conditionsProtective Clothing • leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visorsequipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator	1	2	-	-
PC2. identify job-site hazards and possible causes of risks/accident at the workplaceHazards-electrical hazards; sharp edged and heavy tools; heated metals; oxyfuel and gas cylinders; welding radiation; hazardous surfaces; hazardous substances; physical hazards possible causes of risk and accident: physical actions; not following instructions; inattention; sickness and incapacity; health; not taking safety precautions	1	2	-	-
PC3. follow procedures for safe working with electrical equipmentSafe Working Procedures: tag out/lock out, ptw (permit to work)	1	2	-	-
PC4. follow warning signs while working with electrical systems Warning Signs: danger, high voltage area, out of service, etc	1	2	-	-
PC5. use standard safe working practices when working at heights, confined areas and trenches	1	2	-	-
PC6. test any electrical equipment and system using insulated testing devices before touching them	1	2	-	-
PC7. ensure positive isolation of electrical equipment & system as per the given standards	1	1	-	-
PC8. identify any abnormalities in electrical equipment or system installed alarm and/or noticing parameters from gauge/ indicator installedParameters: temperature, pressure, flow and current	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. follow safe working practices while working in hazardous conditions to ensure the safety of self and others Safe Working Practices: using ppe; installing and reading safety signs; handle tools in the correct manner and store and maintain them properly; keep work area clear of clutter; while working with electricity take all electrical precautions; safe lifting and carrying practices; use equipment that is working properly and is well maintained; take due measures for safety while working at heights, etc. take due measures for safety while working in confined spaces or trenches, etc.	1	2	-	-
PC10. use various methods of accident prevention in the work environmentMethods of Accident Prevention: training in health and safety procedures; using health and safety procedures; use of equipment and working practices; safety notices, advice; instruction from colleagues and supervisors	1	2	-	-
PC11. identify and retrieve general health and safety equipment in the workplaceGeneral Health and Safety Equipment: fire extinguishers; first aid equipment; safety instruments and clothing etc.	1	2	-	-
PC12. set up and use scaffolds, elevated platforms and ladders safelySet up: firm/level base, clip/lash down, leaning at the correct angle, appropriate load as per capacity, etc	1	3	-	-
PC13. inspect scaffolds, elevated platforms and ladders for faults and rectify themFaults: corrosion of metal components, deterioration, splits and cracks in timber components, imbalance, loose rungs, missing/ unfixed nuts or bolts, etc.	1	3	-	-
PC14. lift, carry and transport heavy objects & tools safely using correct procedures from storage to the workplace and vice versa	-	3	-	-
PC15. inspect power plant and its equipment routinely for any signs of oil, water and steam leakage	1	3	-	-
PC16. store flammable materials and machine lubricating oil safely and correctly	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC17. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	2	-	-
PC18. use good housekeeping practices at all timesGood Housekee Practices: clean/tidy work area, removal/disposal of waste material, protect surfaces	1	2	-	-
PC19. identify common hazard signs displayed in various areasVarious Areas: chemical containers; equipment; packages; inside buildings; open areas and public spaces, etc.	1	2	-	-
PC20. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay	1	2	-	-
Follow fire safety practices	**3**	**8**	**-**	**-**
PC21. use various appropriate fire extinguishers for different types of fires correctlyTypes of Fires: class a, b, c, d and e	1	2	-	-
PC22. use appropriate rescue techniques during fire hazard	1	2	-	-
PC23. maintain good housekeeping in order to prevent fire hazards	1	2	-	-
PC24. use a fire extinguisher correctly in case of fire	-	2	-	-
Follow emergencies, rescue and first-aid procedures	**8**	**19**	**-**	**-**
PC25. administer appropriate first aid to the victims for the relevant injuriesInjuries: bleeding, burns, choking, electric shock, poisoning etc.	1	2	-	-
PC26. act promptly and appropriately to an accident situation or medical emergency in real or simulated environments	1	2	-	-
PC27. perform and organise loss minimization or rescue activity during an accident in real or simulated environments	1	4	-	-
PC28. administer basic first aid including cpr to the victims of heart or cardiac arrest due to electric shock, before the arrival of emergency services	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. demonstrate basic emergency procedures during emergency mock drills for better understanding of procedures applied in emergenciesEmergency Procedures: raising alarm, safe/efficient evacuation, correct ways of escape, correct assembly point, roll call, correct return to work	1	2	-	-
PC30. free an affected person from electrocution safely as per the recommended procedure	1	2	-	-
PC31. move injured people and others to a safe place during emergencies	-	3	-	-
PC32. write an incident report or dictate a report to another person, and send report to the authorized personIncident Report Includes Details of: name, date/time of incident, date/time of report, location, environment conditions, persons involved, sequence of events, injuries sustained, damage sustained, actions taken, witnesses, supervisor/manager notified	2	2	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1331
NOS Name	Apply basic health and safety practices for power related work
Sector	Power
Sub-Sector	Distribution, Transmission, Generation, Distribution Downstream
Occupation	Operation & Maintenance - Generation Supervision - Distribution Erection, Installation and Commissioning - DistributionSurveying - TransmissionOperation & Maintenance - Transmission
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	11/02/2019
Next Review Date	11/02/2023
NSQC Clearance Date	22/08/2019

PSS/N1336: Working effectively with others

Description

This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behavior and interactions with others at the workplace. These cover areas such as communication etiquette,discipline, listening, handling conflict and grievances.

Elements and Performance Criteria

working with others

To be competent, the user/individual on the job must be able to:

PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required

PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt

PC3. give information to others clearly, at a pace and in a manner that helps them to understand

PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible

PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks

PC6. display appropriate communication etiquette while working

PC7. display active listening skills while interacting with others at work

PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism

PC9. demonstrate responsible and disciplined behaviors at the workplace

PC10. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions

KU2. reporting structure, inter-dependent functions, lines and procedures in the work area

KU3. relevant people and their responsibilities within the work area

KU4. eescalation matrix and procedures for reporting work and employment related issues

KU5. various categories of people that one is required to communicate and co-ordinate with in the organization

KU6. importance of effective communication in the workplace

KU7. importance of teamwork in organizational and individual success

KU8. various components of effective communication

KU9. key elements of active listening

KU10. value and importance of active listening and assertive communication

KU11. barriers to effective communication
KU12. importance of tone and pitch in effective communication
KU13. importance of avoiding casual expletives and unpleasant terms while communicating professional circles
KU14. how poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer
KU15. importance of ethics for professional success
KU16. importance of discipline for professional success
KU17. what constitutes disciplined behavior for a working professional
KU18. common reasons for interpersonal conflict
KU19. importance of developing effective working relationships for professional success
KU20. expressing and addressing grievances appropriately and effectively
KU21. importance and ways of managing interpersonal conflict effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. note the information communicated by the officer incharge
GS2. note down observations (if any) related to the operation/maintenance
GS3. read and interpret the process required for different types of manuals
GS4. read and interpret the flowchart of all parts of an assembly
GS5. read manuals and documents to understand the product-details & how they can be used
GS6. discuss task lists, schedules and activities with the colleague/supervisor
GS7. effectively communicate with the team members
GS8. attentively listen and comprehend the information given by the colleague/supervisor/contractor
GS9. communicate clearly with the colleague on the issues faced during query/fault
GS10. follow colleague/contractor rule-based decision making process
GS11. take decisions with systematic course of actions and/or response
GS12. planning and organization of tasks to meet deadlines
GS13. build customer relationships and use customer centric approach
GS14. seek and comprehend operation related inputs for clarification find ways of modifying difficult operating stages to make it operation friendly
GS15. work systematically and logically to resolve the issues and identify causation and anticipate unexpected results. Quick approach and solution towards faults repairing
GS16. critically evaluate operation parameters in relation to system normality develop a holistic and comprehensive profile of grid station on segregated discrete processes

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
working with others	**30**	**70**	**-**	**-**
PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	3	7	-	-
PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	3	7	-	-
PC3. give information to others clearly, at a pace and in a manner that helps them to understand	3	7	-	-
PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible	3	7	-	-
PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	3	7	-	-
PC6. display appropriate communication etiquette while working	3	7	-	-
PC7. display active listening skills while interacting with others at work	3	7	-	-
PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	3	7	-	-
PC9. demonstrate responsible and disciplined behaviors at the workplace	3	7	-	-
PC10. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	3	7	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	31/03/2025
NSQC Clearance Date	01/03/2023

PSS/N3401: Perform repair and maintenance of power transmission lines and components

Description

This unit is about the competencies required for repair and maintenance of sub-stations and power transmission lines.

Scope

This unit/ task covers the following:

- Prepare for repair and maintenance of Power Transmission lines
- Perform maintenance of Power Transmission lines
- Perform post repair and maintenance activities

Elements and Performance Criteria

Prepare for repair and maintenance of Power Transmission lines

To be competent, the user/individual on the job must be able to:

PC1. identify various types of circuits and their components correctly

PC2. identify and acquire correct tools, equipment and instruments required for various aspects of repair, maintenance, assessment and inspection of transmission lines and components

PC3. identify hazards in trimming trees such as limits of approach, public safety and step and touch potential access

PC4. conduct site inspection for emergency cases following established procedures

PC5. climb tower while following all necessary safety procedures and using ppe

PC6. identify various types of circuits accurately

PC7. ensure the tools and equipment are well maintained, calibrated and approved for use

PC8. use transmission line tools, equipment and hardware in line with job requirements for maintenance operations

PC9. prepare and maintain the work area as per procedure or operation specification

PC10. obtain work permit to proceed with work from the appropriate personnel in accordance with the organizational procedure

PC11. switch off, isolate, discharge and earth (side) line cables

Perform maintenance of power transmission lines as job specifications and requirements

To be competent, the user/individual on the job must be able to:

PC12. perform off-line overhead line maintenance procedure

PC13. perform off-line underground line maintenance procedure

PC14. ensure pole dismantling and re-setting procedure is carried out as per standard procedure, where required

PC15. install components on transmission lines including gang operated air brake switches for transmission lines, controlled breakers, ground switches,capacitor stations, insulator pressure washing, submarine and underground transmission cable, grid interconnections, etc.

PC16. select and use test equipment such as tong testers, clip-on meter, multimeters, fault indicators meggers and voltmeters to verify fault and integrity

PC17. document switching procedures with all relevant details clearly and accurately

PC18. repair conductor by splicing, jointing, using armor rods, line guards, vibration dampers

PC19. evaluate the work carried out by the team members and ensure it is as per the standard requirements and provide any feedback in a timely, polite and supportive manner

PC20. report any issues along with the remedial actions such as repairs or replacements, and estimated repair time to the relevant authority

PC21. replace pole as per standard procedure, where required

PC22. carry out guy and anchor replacement on various structure typesstructure types: wood, steel, various lines voltages

PC23. carry out conductor patch and splice repair on single conductor, bundled conductor of various sizes and line voltages

PC24. replace components such as transformers, ct (current transformer), cvt (capacitive voltage transformer), la (lightening arrestor), breakers, towers, conductors, disconnects, timber or x-arm, conductors, poles, switches, elbows, terminations and insulators safely

PC25. replace other line components due to damage or unsuitability as per standard procedure, where required

PC26. replace underground cables, as per standard procedures where required

Perform post repair and maintenance activities

To be competent, the user/individual on the job must be able to:

PC27. restore the system to normal operating status by using switching procedures

PC28. identify problems that are not within one's own capability or limits of authority and seek support from authorized persons for resolving the same

PC29. leave the work area in a safe and tidy condition on completion of the repair and maintenance activities

PC30. refer unresolved job-related problems to the appropriate personnel for support

PC31. monitor the problem and keep the supervisor informed about progress or any delays in resolving the problem

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. relevant legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
2. relevant health and safety requirements applicable in the work place
3. own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
4. reporting structure, inter-dependent functions, lines and procedures in the work area

5. how to engage with specialists for support in order to resolve incidents and service requests
6. importance of working in clean and safe environment practices and procedures
7. relevant people and their responsibilities within the work area
8. escalation matrix and procedures for reporting work and employment related issues
9. principles of electricity
10. common electricity terminology and correct interpretation of the same
11. elements of the power system elements: generation, transmission, transmission metering, etc.elements: generation, transmission, transmission metering, etc.
12. different types of material and accessories used in power t&d materials and accessories: support (towers-steel, cement, wooden),materials and accessories: support (towers-steel, cement, wooden),
13. sizes and current carrying capacity of different types of conductors
14. conductor accessories: binding tape, binding wire, p.g. clamp, t clamp etc.; insulators - pin, disc, shackle, guy etc., lt switchgears, cross arms, stay sets, go switches etc. types of cross arms, etc.
15. tools and equipment: plier, screwdriver, wrench set, hammer, drilling machine, hacksaw/cutting tools, measuring tape, pulleys (force pulley with sling), tommy bar, crimping machine, round / flat file, earth rod i.e. discharge rod), leakage current monitoring kit
16. specific health and safety precautions which must be taken when carrying out transmission lines repair and maintenance work especially live line orequipment
17. various types of circuits and equipmenttypes: ct, pt, control, indication & annunciation circuits
18. line diagrams, maps and circuitry
19. troubleshooting and repair methods for various components of transmission lines
20. fault indicators for various components of the transmission lines
21. overhead transmission system apparatus such as regulators and reclosers
22. overhead transmission system standards
23. access points such as vaults, open trenches and manholes
24. underground transmission system apparatus such as transformers, switching cables and junction boxes
25. cable locating and fault detecting equipment
26. co-existing underground utilities
27. types and sizes of conductors and cables
28. causes of conductor damage
29. classification of conductor and insulator damage including fretting, abrasion,
30. fatigue breaks, tensile breaks
31. need for an authorized permit on 11 kv and above voltage line, if the line is under own authority, then take self-permit
32. hazards associated with carrying out power line maintenance and how they can be minimized
33. importance of ensuring that tools and equipment are suitable, well maintained, calibrated and operating effectively
34. importance of following good housekeeping and fire prevention procedures
35. importance of following job instructions and defined maintenance procedures

36. material preparation methods and techniques to be undertaken, prior to using for testing and maintenance activities
37. preparation of equipment for testing and repair activities
38. components of transmission lines
39. procedures for handling transmission line components with imperfections/defects that cannot be removed/repaired and how can they be minimized
40. problems and conditions which render electrical poles or towers in need of maintenance or replacement
41. importance of leaving the work area and equipment in a safe and clean condition on completion of the repair and maintenance activities
42. importance of reporting problems in a timely manner
43. methods and parameters to check quality of line components against required quality standards methods: visual inspection, binoculars, measuring tape, use of instruments
44. calibration schedule of all equipment used in inspection, repair and maintenance activities
45. principles and practices of electrical safety
46. standard procedures how to deal with electric shocks and electrocutions to rescue and minimize damage and harm
47. safety concerns and precautions for working at significant height and with high wind velocities
48. Personal protective equipment (ppe) and clothing that must be worn during The inspection, repair and maintenance activity and from where can it be Obtained

Generic Skills (GS)

User/individual on the job needs to know how to:

1. write official notes
2. write the information communicated by the work in-charge
3. write about the technical problems and other conditions of the site as per standards
4. note critical points and testing repair observations
5. write about the condition of equipment
6. fill in all technical forms and data as per guidelines and format
7. read and correctly interpret written sentences and paragraphs
8. read and correctly interpret units of metric system for all measurements
9. read and correctly interpret the process required for performing work
10. read and correctly interpret the rules and methods from policy and procedure documents
11. read and correctly interpret equipment manuals to identify operational details and specifications
12. communicate clearly with team and other staff members
13. discuss task lists, schedules and activities
14. interact with the junior engineer to clarify instructions and doubts
15. make appropriate work-related judgments based on policy and procedures
16. identify complex problems and review related information to develop and evaluate solutions

17. follow organizations rule-based decision-making process
18. plan and organize tasks to meet quantity and quality specifications and targets
19. build customer relationships and use customer centric approach in taking decisions
20. identify problems and review related information to develop and evaluateoptions and implement solutions
21. monitor problem solving to take corrective action with individuals andorganizations
22. analyse problems and changes in conditions, operations, and the environment to solve problems
23. analyse the problems in the equipment to identify root causes
24. collect information and technical data and define process for equipment testing
25. analyse the problem and required actions to resolve them to determine which issues are to be escalated, and to whom
26. develop holistic and comprehensive profile of products based on segregated discrete process stages
27. critically evaluate operation parameters in relation to product features intended to ensure a match

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Prepare for repair and maintenance of Power Transmission lines	**8**	**25**	**-**	**-**
PC1. identify various types of circuits and their components correctly	1	3	-	-
PC2. identify and acquire correct tools, equipment and instruments required for various aspects of repair, maintenance, assessment and inspection of transmission lines and components	1	2	-	-
PC3. identify hazards in trimming trees such as limits of approach, public safety and step and touch potential access	1	2	-	-
PC4. conduct site inspection for emergency cases following established procedures	1	2	-	-
PC5. climb tower while following all necessary safety procedures and using ppe	-	2	-	-
PC6. identify various types of circuits accurately	1	3	-	-
PC7. ensure the tools and equipment are well maintained, calibrated and approved for use	-	2	-	-
PC8. use transmission line tools, equipment and hardware in line with job requirements for maintenance operations	1	2	-	-
PC9. prepare and maintain the work area as per procedure or operation specification	1	3	-	-
PC10. obtain work permit to proceed with work from the appropriate personnel in accordance with the organizational procedure	1	2	-	-
PC11. switch off, isolate, discharge and earth (side) line cables	-	2	-	-
Perform maintenance of power transmission lines as job specifications and requirements	**17**	**34**	**-**	**-**
PC12. perform off-line overhead line maintenance procedure	1	3	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. perform off-line underground line maintenance procedure	1	2	-	-
PC14. ensure pole dismantling and re-setting procedure is carried out as per standard procedure, where required	1	2	-	-
PC15. install components on transmission lines including gang operated air brake switches for transmission lines, controlled breakers, ground switches,capacitor stations, insulator pressure washing, submarine and underground transmission cable, grid interconnections, etc.	2	3	-	-
PC16. select and use test equipment such as tong testers, clip-on meter, multimeters, fault indicators meggers and voltmeters to verify fault and integrity	1	3	-	-
PC17. document switching procedures with all relevant details clearly and accurately	2	2	-	-
PC18. repair conductor by splicing, jointing, using armor rods, line guards, vibration dampers	1	2	-	-
PC19. evaluate the work carried out by the team members and ensure it is as per the standard requirements and provide any feedback in a timely, polite and supportive manner	1	2	-	-
PC20. report any issues along with the remedial actions such as repairs or replacements, and estimated repair time to the relevant authority	1	2	-	-
PC21. replace pole as per standard procedure, where required	1	2	-	-
PC22. carry out guy and anchor replacement on various structure typesstructure types: wood, steel, various lines voltages	1	2	-	-
PC23. carry out conductor patch and splice repair on single conductor, bundled conductor of various sizes and line voltages	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. replace components such as transformers, ct (current transformer), cvt (capacitive voltage transformer), la (lightening arrestor), breakers, towers, conductors, disconnects, timber or x-arm, conductors, poles, switches, elbows, terminations and insulators safely	1	3	-	-
PC25. replace other line components due to damage or unsuitability as per standard procedure, where required	1	2	-	-
PC26. replace underground cables, as per standard procedures where required	1	2	-	-
Perform post repair and maintenance activities	**5**	**11**	**-**	**-**
PC27. restore the system to normal operating status by using switching procedures	1	2	-	-
PC28. identify problems that are not within one's own capability or limits of authority and seek support from authorized persons for resolving the same	1	2	-	-
PC29. leave the work area in a safe and tidy condition on completion of the repair and maintenance activities	-	2	-	-
PC30. refer unresolved job-related problems to the appropriate personnel for support	2	2	-	-
PC31. monitor the problem and keep the supervisor informed about progress or any delays in resolving the problem	1	3	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N3401
NOS Name	Perform repair and maintenance of power transmission lines and components
Sector	Power
Sub-Sector	Transmission
Occupation	Operation & Maintenance - Transmission
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	11/03/2019
Next Review Date	11/03/2023
NSQC Clearance Date	22/08/2019

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1.Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.

2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.

3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).

4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.

5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.

6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N1331.Apply basic health and safety practices for power related work	30	70	-	-	100	15
PSS/N1336.Working effectively with others	30	70	-	-	100	5
PSS/N3401.Perform repair and maintenance of power transmission lines and components	30	70	-	-	100	80
Total	**90**	**210**	**-**	**-**	**300**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
CPR	Cardiopulmonary Resuscitation
PPE	Personal Protective Equipment
CT	Current Transformer
PT	Potential Transformer
CVT	Capacitive Voltage Transformer
LA	Lightening Arrestor
CB	Circuit Breaker
AC	Alternating Current
DC	Direct Current

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.